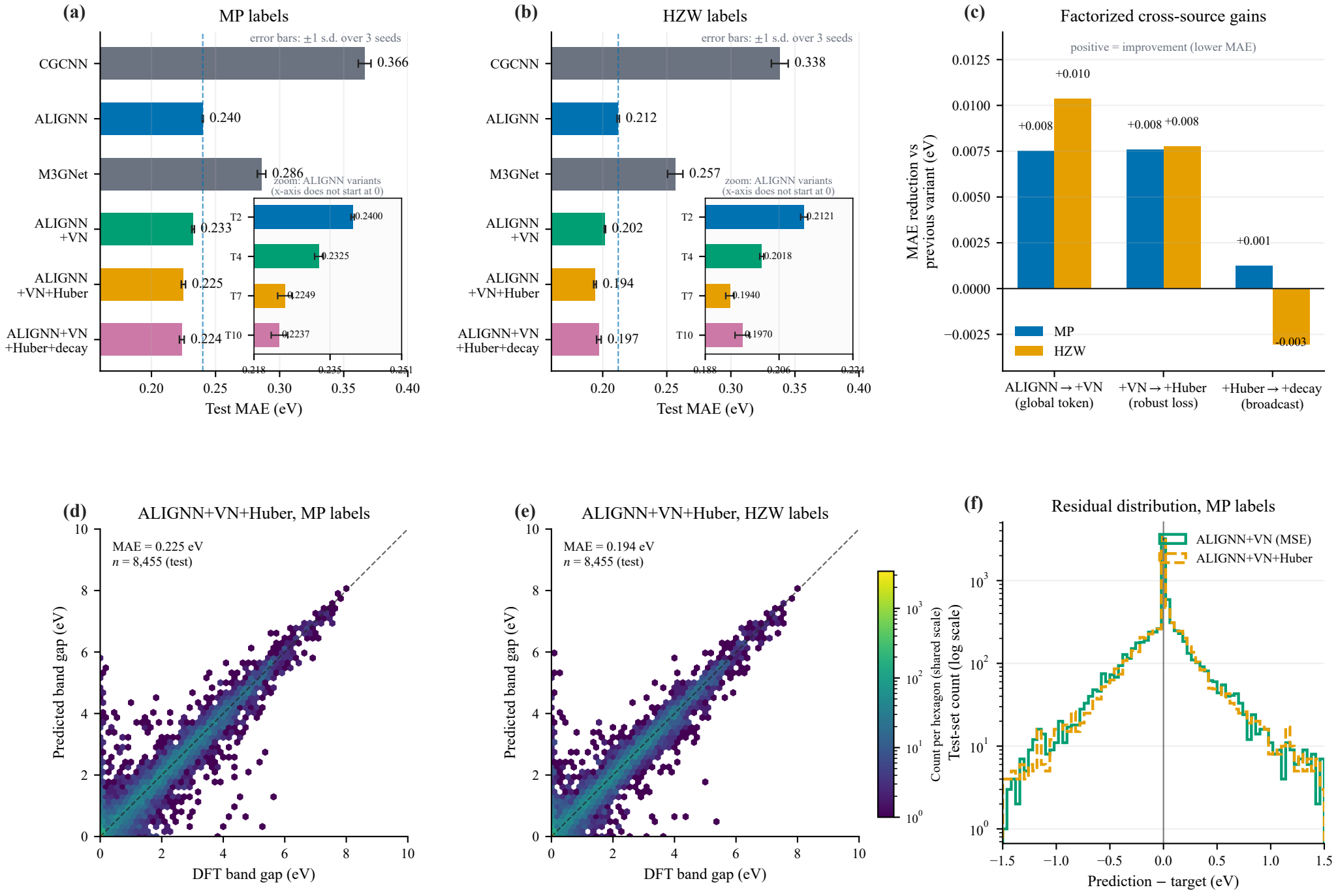

(a)
MP labels
error bars: ±1 s.d. over 3 seeds
CGCNN
0.366
ALIGNN
0.240
M3GNet
0.286
zoom: ALIGNN variants
(x-axis does not start at 0)
ALIGNN +VN
0.233
T2
0.2400
T4
0.2325
ALIGNN +VN+Huber
0.225
T7
0.2249
ALIGNN+VN +Huber+decay
0.224
T10
0.2237
0.218
0.235
0.251
0.20
0.25
0.30
0.35
0.40
Test MAE (eV)
(b)
HZW labels
error bars: ±1 s.d. over 3 seeds
CGCNN
0.338
ALIGNN
0.212
M3GNet
0.257
zoom: ALIGNN variants
(x-axis does not start at 0)
ALIGNN +VN
0.202
T2
0.2121
T4
0.2018
ALIGNN +VN+Huber
0.194
T7
0.1940
ALIGNN+VN +Huber+decay
0.197
T10
0.1970
0.188
0.206
0.224
0.20
0.25
0.30
0.35
0.40
Test MAE (eV)
(c)
Factorized cross-source gains
positive = improvement (lower MAE)
MAE reduction vs previous variant (eV)
0.0125
0.0100
0.0075
0.0050
0.0025
0.0000
−0.0025
+0.008
+0.010
+0.008
+0.008
+0.001
-0.003
MP
HZW
ALIGNN → +VN (global token)
+VN → +Huber (robust loss)
+Huber → +decay (broadcast)
(d)
ALIGNN+VN+Huber, MP labels
MAE = 0.225 eV
n = 8,455 (test)
Predicted band gap (eV)
DFT band gap (eV)
0
2
4
6
8
10
(e)
ALIGNN+VN+Huber, HZW labels
MAE = 0.194 eV
n = 8,455 (test)
Predicted band gap (eV)
DFT band gap (eV)
Count per hexagon (shared scale)
10^0
10^1
10^2
10^3
(f)
Residual distribution, MP labels
ALIGNN+VN (MSE)
ALIGNN+VN+Huber
Test-set count (log scale)
10^0
10^1
10^2
10^3
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Prediction − target (eV)